



Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900

Purchases

18

Features

\$59.76

Avg. Spend

3.75

Avg. Rating

Demographics

Age, Gender, Location,
Subscription

Purchase Details

Category, Amount, Season, Size,
Color

Behavior

Discounts, Frequency, Ratings,
Shipping

Data Preparation in Python

01

Load & Explore

pandas import, `df.info()` and `.describe()`

02

Clean & Impute

37 missing Review Ratings filled with category median

03

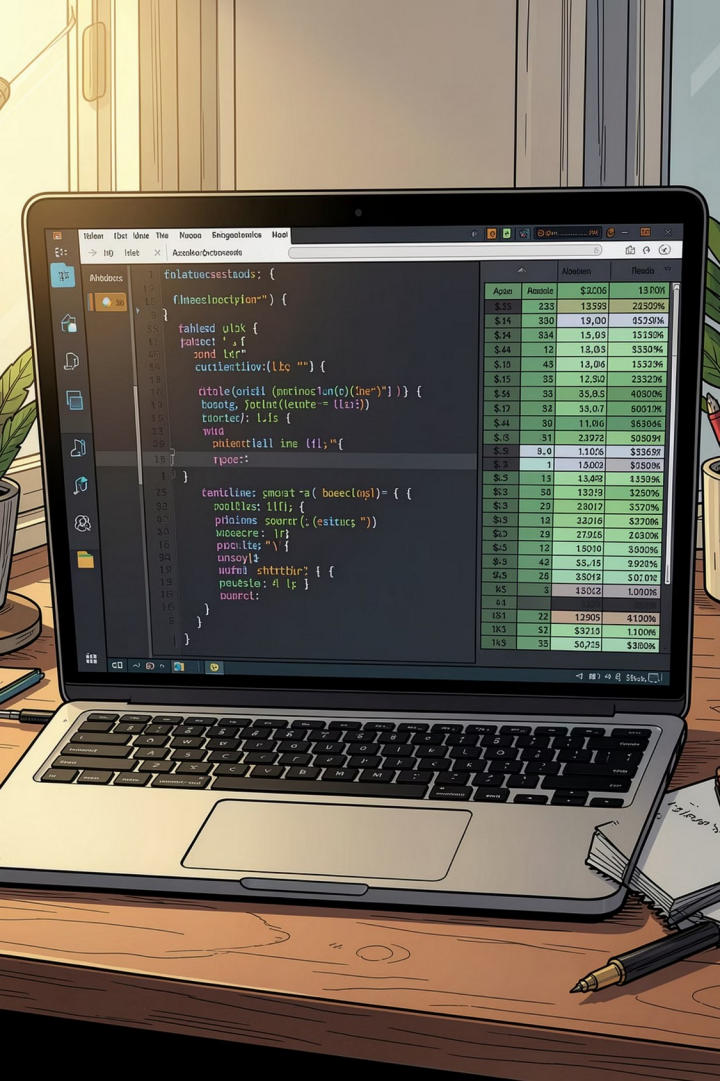
Feature Engineering

Added `age_group` and `purchase_frequency_days` columns

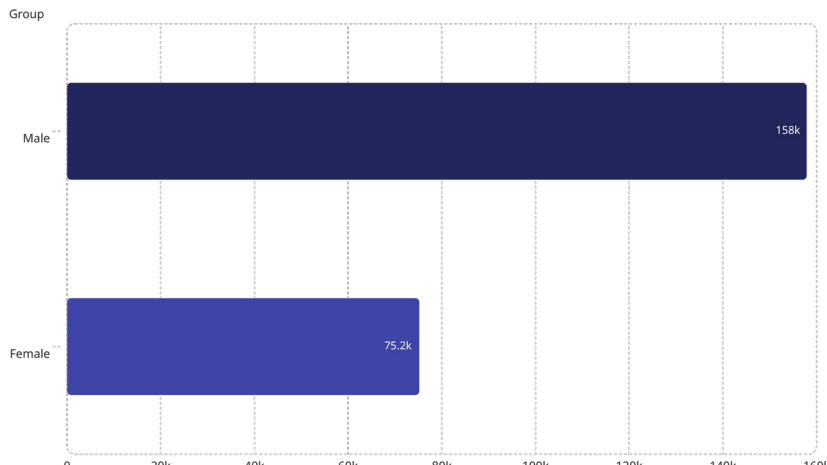
04

Load to SQL

Cleaned DataFrame pushed to MySQL for structured analysis



Revenue & Spending Insights



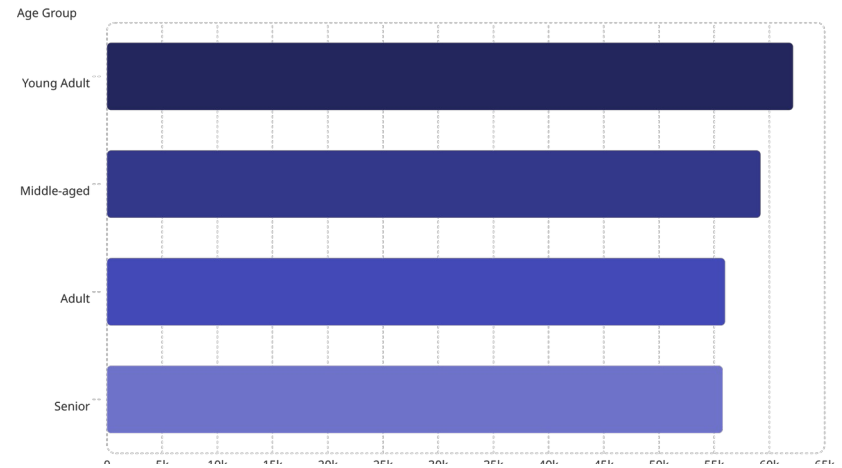
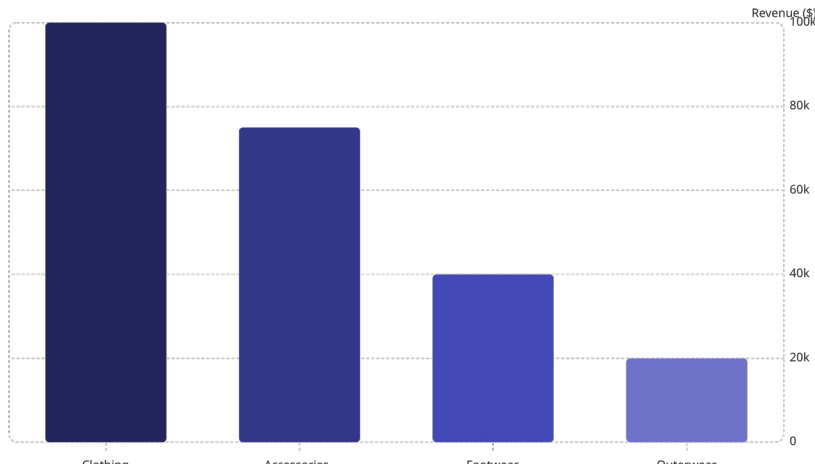
Key Findings

Male customers generate **2× more revenue** than female customers.

Express shipping users spend slightly more (**\$60.48**) vs. Standard (**\$58.46**).

839 discount users still spent above the average purchase amount.

Revenue by Category & Age Group



Clothing leads all categories. Young Adults are the top revenue-generating age group.

Top Products & Discount Behavior

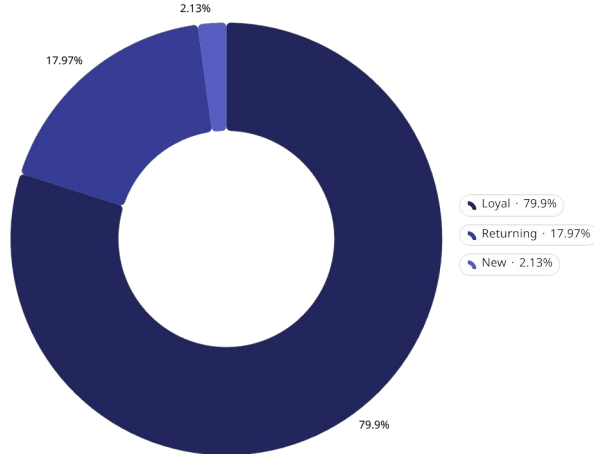
Top 5 by Rating

Product	Avg Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation



Repeat Buyers & Subscriptions

Of customers with >5 purchases:

Loyal customers dominate — yet most repeat buyers remain **unsubscribed**.

2,518

Not subscribed

958

Subscribed

Subscriber vs. Non-Subscriber

Subscribers

1,053 customers

Avg Spend: **\$59.49**

Total Revenue: **\$62,645**

Non-Subscribers

2,847 customers

Avg Spend: **\$59.87**

Total Revenue: **\$170,436**

 Non-subscribers account for **73%** of customers and **73%** of total revenue — a major growth opportunity.

Power BI Dashboard

Interactive dashboard with filters for Subscription, Gender, Category, and Shipping Type — surfacing all key metrics in one view.



Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base

Review Discount Policy

Balance sales boosts with margin control on high-discount products

Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment

Targeted Marketing

Focus on Young Adults and express-shipping users for highest ROI